

B.TECH • SEMESTER I • ENGINEERING PHYSICS

Unit III

Dielectric & Magnetic Materials

Complete Formula + Concept Revision Sheet



Dielectric Materials



Magnetic Materials



Clausius-Mossotti



Hysteresis Loop



All Key Formulas



25 Flashcards

- Polarization Types
- Complex Dielectric Constant
- Atomic Origin of Magnetism
- Ferromagnetism & Domains
- Lorentz Internal Field
- Frequency Dependence
- Magnetic Classification
- Soft vs Hard Magnets

§1 Dielectric Materials – Core Concepts

Dielectric Material: An insulating material that, when placed in an electric field, does not conduct current but gets *polarized*. Bound charges shift slightly to create dipoles. Examples: glass, mica, water, ceramics.

Electric Polarization (P): Net dipole moment per unit volume of the dielectric.

Unit: C/m² | Vector quantity (along E)

Dielectric Polarizability (α): Proportionality constant between induced dipole moment and applied field for a single atom/molecule.

$$p = \alpha \epsilon_0 E \quad \text{or} \quad p = \alpha E$$

Dielectric Constant (ε_r): Ratio of permittivity of material to permittivity of free space.

$$\epsilon_r = \epsilon / \epsilon_0 \geq 1 \quad (\text{dimensionless})$$

⚡ Fundamental Relations Between D, E, P

$$\begin{aligned} D &= \epsilon_0 E + P && [\text{Displacement vector}] \\ P &= \epsilon_0 \chi_e E && [\text{Polarization vs field}] \\ D &= \epsilon_0 \epsilon_r E && [\text{Alternate form}] \\ \epsilon_r &= 1 + \chi_e && [\text{Key relation}] \end{aligned}$$

Symbol	Quantity	Unit	Description
D	Electric Displacement	C/m ²	Total charge density effect (free + bound)
E	Electric Field Intensity	V/m	Applied external field
P	Polarization	C/m ²	Dipole moment per unit volume
ε ₀	Permittivity of free space	F/m	8.85 × 10 ⁻¹² F/m
ε _r	Relative permittivity	–	Dimensionless, ≥ 1
χ _e	Electric susceptibility	–	Dimensionless; = ε _r – 1
α	Polarizability	C·m / (V/m)	Per atom/molecule response

📊 Derived Relations (Quick Reference)

$$\begin{aligned} P &= N \alpha \epsilon_0 E \quad (N = \text{number density}) \\ P &= (\epsilon_r - 1) \epsilon_0 E \\ \chi_e &= \epsilon_r - 1 \\ \epsilon &= \epsilon_0 \epsilon_r = \epsilon_0 (1 + \chi_e) \end{aligned}$$

$$\begin{aligned} D &= \epsilon_0 E + P = \epsilon_0 \epsilon_r E \\ D &= \epsilon E \quad (\text{where } \epsilon = \epsilon_0 \epsilon_r) \\ P / \epsilon_0 E &= \epsilon_r - 1 = \chi_e \end{aligned}$$

⚠ Common Error: Students confuse ε_r (relative permittivity/dielectric constant) with χ_e (susceptibility). Remember: ε_r = 1 + χ_e. For vacuum, χ_e = 0 and ε_r = 1.

§2 Types of Polarization

① Electronic Polarization

- Electron cloud shifts with respect to nucleus under applied E field.
- Present in *all* materials; fastest mechanism (occurs at optical frequencies).

Induced dipole moment: $p_e = \alpha_e \cdot E$

- ♦ Electronic polarizability: $\alpha_e = 4\pi\epsilon_0 R^3$ (R = atomic radius)
- ♦ Temperature independent

② Ionic Polarization

- ♦ Positive and negative ions in ionic crystals (e.g., NaCl) displace relative to each other.
- ♦ Induced dipole moment: $p_i = \alpha_i \cdot E$
- ♦ Ionic polarizability: $\alpha_i = \frac{e^2}{\omega_0^2} \cdot \left(\frac{1}{M^+} + \frac{1}{M^-}\right)$ (ω_0 = resonant freq.)
- ♦ Slower than electronic; Temperature independent (approximately)

③ Orientation Polarization (Debye)

- ♦ Molecules with permanent dipoles (e.g., H₂O, HCl) rotate and align with E.
- ♦ Orientation polarizability: $\alpha_{or} = \frac{p_0^2}{3k_B T}$
- ♦ **Temperature dependent** — increases as temperature decreases.

④ Space Charge Polarization (Interface Polarization)

- ♦ Charges accumulate at grain boundaries / interfaces in inhomogeneous materials.
- ♦ Slowest mechanism; only at very low frequencies.

Polarization Types – Comparison Table

Type	Mechanism	Frequency Range	Temp. Dep.	Examples
Electronic	Electron cloud displacement	UV / Optical ($\sim 10^{15}$ Hz)	Independent	All materials
Ionic	Ion pair displacement	IR ($\sim 10^{13}$ Hz)	Independent	NaCl, KCl, MgO
Orientation	Permanent dipole alignment	Microwave ($\sim 10^{10}$ Hz)	Decreases with T	H ₂ O, HCl, CH ₃ Cl
Space Charge	Interface charge accumulation	Very low ($\sim 10^2$ Hz)	Dependent	Ceramics, Polymers

💡 **Memory Aid:** E.I.O.S → *Every Ion Orients Slowly* (Electronic → Ionic → Orientation → Space charge, from fastest to slowest)

§3 Lorentz Internal Field

- ♦ The local (internal) electric field acting on an atom/molecule in a dielectric is *not* the same as the applied field E .
- ♦ The local field is larger than E due to contributions from surrounding dipoles.

$$E_{\text{local}} = E_{\text{int}} = E + P / (3\epsilon_0) \quad [\text{Lorentz Internal Field}]$$

$$E_{\text{local}} = (\epsilon_r + 2)/3 \cdot E \quad [\text{For linear dielectrics}]$$

COMPONENTS OF E_{LOCAL}

- ♦ E_1 = Field from external source
- ♦ E_2 = Depolarization field
- ♦ E_3 = Lorentz cavity field = $P/3\epsilon_0$
- ♦ E_4 = Field from dipoles inside sphere ≈ 0 (cubic)

PHYSICAL MEANING

- ♦ Accounts for mutual interactions of dipoles
- ♦ The cavity is a Lorentz sphere around each atom
- ♦ $P/3\epsilon_0$ is the contribution of all surrounding dipoles

§4 Clausius–Mossotti Equation

- ♦ Relates macroscopic quantity (ϵ_r) to microscopic quantity (α) using the Lorentz internal field.

$$(\epsilon_r - 1) / (\epsilon_r + 2) = N \alpha / (3\epsilon_0)$$

Symbol	Meaning	Unit
ϵ_r	Relative permittivity (dielectric constant)	Dimensionless
N	Number of atoms/molecules per unit volume	m^{-3}
α	Polarizability per molecule	$\text{C}\cdot\text{m}^2/\text{V}$ (or $\text{F}\cdot\text{m}^2$)
ϵ_0	Permittivity of free space	$8.85 \times 10^{-12} \text{ F/m}$

EXTENDED FORM (MULTIPLE SPECIES)

$$(\epsilon_r - 1)/(\epsilon_r + 2) = (1/3\epsilon_0) \sum_i N_i \alpha_i$$

OPTICAL FORM (N = REFRACTIVE INDEX)

$$(n^2 - 1)/(n^2 + 2) = N\alpha/(3\epsilon_0)$$

[Since $\epsilon_r = n^2$ at optical freq.]

Application: Used to calculate dielectric constant from molecular polarizability and vice versa. Bridge between atomic properties and bulk behavior.

§5

Complex Dielectric Constant & Dielectric Loss

$\epsilon^* = \epsilon' - j\epsilon''$ [Complex permittivity]

$\tan \delta = \epsilon'' / \epsilon'$ [Loss tangent / dissipation factor]

Component	Symbol	Physical Meaning
Real part	ϵ'	Energy <i>stored</i> in the dielectric (reactive component)
Imaginary part	ϵ''	Energy <i>dissipated</i> as heat (loss component)
Loss tangent	$\tan \delta$	Ratio of energy lost to energy stored per cycle

- ♦ Dielectric loss = energy dissipated as heat when a dielectric is placed in an alternating electric field.
- ♦ High $\epsilon'' \rightarrow$ poor insulator (lossy material)
- ♦ Low $\tan \delta \rightarrow$ good capacitor dielectric (e.g., Teflon)
- ♦ Power dissipated per unit volume: $P = \omega \epsilon_0 \epsilon'' E_0^2 / 2$

§6

Frequency Dependence of Polarization

FREQUENCY SPECTRUM OF POLARIZATION

Space Charge
~10² Hz

Orientation
~10⁸–10¹⁰ Hz

Ionic
~10¹³ Hz

Electronic
~10¹⁵ Hz

LOW FREQUENCY → HIGH FREQUENCY

Total polarizability: $\alpha_{\text{total}} = \alpha_e + \alpha_i + \alpha_{\text{or}} + \alpha_{\text{sc}}$

AT LOW FREQUENCY

- All four mechanisms contribute
- ϵ_r is maximum
- Dielectric loss is small
- Space charge & orientation polarization present

AT HIGH FREQUENCY (OPTICAL)

- Only electronic polarization survives
- ϵ_r reduces to n^2 (optical)
- Heavier mechanisms "lag behind" the field
- Peaks in ϵ'' at resonance = absorption

Resonance/Relaxation: When applied frequency $\approx$ natural frequency of a polarization mechanism, that mechanism's contribution drops sharply (resonance absorption). ϵ' decreases stepwise at each critical frequency.

§7

Magnetic Materials – Core Definitions

Magnetic Dipole Moment (m): Current loop of area A carrying current I.
 $m = I \times A$ Unit: A·m²

Magnetic Susceptibility (χ_m): Measure of how easily a material is magnetized.
 $\chi_m = M / H$ (dimensionless)

Magnetization (M): Net magnetic dipole moment per unit volume.
 $M = \chi_m H$ Unit: A/m

Relative Permeability (μ_r): Ratio of permeability of material to free space.
 $\mu_r = \mu / \mu_0 = 1 + \chi_m$

Key Magnetic Relations

$B = \mu_0(H + M)$
 $B = \mu_0\mu_r H = \mu H$
 $M = \chi_m H$
 $\mu_r = 1 + \chi_m$
 $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

[Magnetic flux density]

[Alternate form]

[Magnetization]

[Analogous to $\epsilon_r = 1 + \chi_e$]

[Permeability of free space]

Symbol	Quantity	Unit	Note
B	Magnetic flux density	Tesla (T)	Total field in material
H	Magnetic field intensity	A/m	Applied/free current contribution
M	Magnetization	A/m	Material's response
μ_0	Permeability of free space	H/m	$4\pi \times 10^{-7} \text{ H/m}$
μ_r	Relative permeability	–	Dimensionless
χ_m	Magnetic susceptibility	–	$= \mu_r - 1$

SOURCES OF MAGNETIC MOMENT

- ♦ **Orbital motion** of electrons around nucleus → orbital magnetic moment
- ♦ **Electron spin** (intrinsic angular momentum) → spin magnetic moment
- ♦ **Nuclear spin** (much smaller, $\sim 10^{-3}$ of electron)

BOHR MAGNETON

$$\mu_B = e\hbar / (2m_e) = 9.274 \times 10^{-24} \text{ A} \cdot \text{m}^2$$

- ♦ Basic unit of magnetic moment
- ♦ Orbital m.m. = $m_l \times \mu_B$
- ♦ Spin m.m. = $\pm \mu_B$ (for $s = \pm 1/2$)

§9

Classification of Magnetic Materials

Property	Diamagnetic	Paramagnetic	Ferromagnetic	Antiferro-magnetic	Ferri-magnetic
χ_m value	Small, negative -10 ⁻⁵ to -10 ⁻⁶	Small, positive 10 ⁻⁵ to 10 ⁻³	Large, positive 10 ² to 10 ⁵	Small, positive ~10 ⁻³ to 10 ⁻⁵	Large, positive 10 to 10 ⁴
Dipole alignment	No permanent dipoles; induced opp. to H	Random; weakly align with H	Parallel (same direction)	Antiparallel, equal magnitude → cancel	Antiparallel, unequal → net moment
Behavior in field	Repelled weakly	Attracted weakly	Strongly attracted	Weakly attracted	Attracted (moderate)
Domain structure	No domains	No domains	Domains present	No net domains	Domains present
Temp. dependence	Nearly none	Curie law: $\chi = C/T$	Curie-Weiss: $\chi = C/(T-T_c)$	Becomes para above Néel temp. T _N	Becomes para above T _c (Curie)
Transition temp.	—	—	Curie temp. T _c	Néel temp. T _N	Curie temp. T _c
Examples	Cu, Ag, Au, Bi, water, Hg	Al, Pt, Mn, CuSO ₄	Fe, Ni, Co, Gd	MnO, FeO, Cr	Fe ₃ O ₄ , ferrites, MnZn ferrite
Hysteresis	None	None	Yes (large loop)	None	Yes (small loop)

▢

Temperature Laws (Curie & Néel)

Curie Law (Paramagnetic): $\chi_m = C / T$ (T in Kelvin) C = Curie constant	Curie-Weiss Law (Ferromagnetic): $\chi_m = C / (T - T_c)$ T_c = Curie temperature (K)
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Above T_c: Ferromagnetic → Paramagnetic | Above T_N: Antiferromagnetic → Paramagnetic

§10

Ferromagnetism & Magnetic Domains

Domain: A small region in a ferromagnet where <i>all atomic dipoles are aligned in the same direction</i>. Each domain acts like a small magnet. Size: ~0.001–1 mm.	Weiss Molecular Field: An internal (exchange) field that keeps spins aligned within a domain. $H_{mol} = \gamma M$ (γ = Weiss constant)
Domain Wall (Bloch Wall): The boundary between two adjacent domains. Spins gradually rotate across the wall. Width: ~100–1000 nm.	Exchange Interaction: Quantum mechanical interaction between adjacent electron spins responsible for parallel alignment in ferromagnets. Stronger than classical dipole interactions.

🔄

Magnetization Process Step by Step

UNMAGNETIZED → FULLY MAGNETIZED

Step 1

Domains random
Net $M = 0$



Step 2

Wall motion
Favorable domains grow



Step 3

Domain rotation
toward H direction



Step 4

Saturation
Single domain

§11

Hysteresis Loop (B-H Curve)



Key Parameters of the Loop

Term	Symbol	Definition
Saturation	Bs	Maximum B achievable; all domains aligned
Retentivity	Br	B remaining when H=0 after saturation
Coercivity	Hc	Reverse H needed to reduce B to zero
Remnance	Mr	Magnetization remaining after H removed

Energy loss per cycle = Area of B-H loop
Unit: J/m³/cycle

- Wider loop → more energy loss per cycle (Hard magnet)
- Narrow loop → less energy loss (Soft magnet)
- Hysteresis loss causes heating in transformer cores

§12

Soft vs Hard Magnetic Materials

Property	Soft Magnetic Materials	Hard Magnetic Materials
Coercivity (Hc)	Low (easy to demagnetize)	High (hard to demagnetize)
Retentivity (Br)	Low	High (retains magnetism)
Hysteresis loop	Narrow (small area)	Wide (large area)
Permeability (μr)	Very high	Low to moderate
Energy loss	Low per cycle	High per cycle
Magnetic saturation	Low to moderate	High
Eddy current loss	Lower (high resistivity preferred)	Acceptable (DC applications)
Applications	Transformer cores, inductors, electromagnets, motor cores, shielding	Permanent magnets, magnetic storage, speakers, MRI machines
Examples	Silicon steel (Fe-Si), Permalloy, Ferrites (MnZn), Mumetal	Alnico, SmCo, NdFeB, CrO ₂ , Ba-ferrite
Domain wall motion	Easy (low friction)	Difficult (pinned walls)

💡 **Memory Tip:** SOFT = Small coercivity, Small area loop, Transformer use | HARD = High H_c , Hard to demagnetize, Holds magnetism permanently.

§13 Rapid Recall Flashcards

Q: What is dielectric polarization (P)?

P = Dipole moment per unit volume (C/m^2)

Q: What is ϵ_r in terms of χ_e ?

$\epsilon_r = 1 + \chi_e$

Q: What is the Clausius–Mossotti equation?

$(\epsilon_r - 1)/(\epsilon_r + 2) = Na/(3\epsilon_0)$

Q: Which polarization is temperature dependent?

Orientation (Debye) polarization; $\alpha_{or} = p_o^2/3k_{BT}$

Q: What does ϵ'' (imaginary part) represent?

Energy dissipated (dielectric loss) per cycle

Q: Relation between B, H, M?

$B = \mu_0(H + M)$

Q: What is the Bohr magneton?

$\mu_B = e\hbar/2m_e = 9.274 \times 10^{-24} \text{ A} \cdot \text{m}^2$

Q: Curie law for paramagnetics?

$\chi_m = C/T$ (C = Curie constant)

Q: What is retentivity (Br)?

B remaining in material when H is reduced to zero

Q: What type of material has narrow B-H loop?

Soft magnetic material (e.g., Silicon steel)

Q: What is a magnetic domain?

Region of $\sim 10^{-5}$ – 10^{-6} m^3 with all dipoles aligned parallel

Q: Electronic polarizability in terms of radius R?

$\alpha_e = 4\pi\epsilon_0 R^3$

Q: Hysteresis loss per cycle equals?

Area enclosed by the B-H loop (in J/m^3)

Q: Relation between D, E, P?

$D = \epsilon_0 E + P$

Q: Unit of electric susceptibility χ_e ?

Dimensionless (no units)

Q: What is the Lorentz internal field?

$E_{\text{local}} = E + P/(3\epsilon_0)$

Q: Which polarization is fastest (highest frequency)?

Electronic polarization ($\sim 10^{15} \text{ Hz}$, UV/optical)

Q: What is the loss tangent?

$\tan \delta = \epsilon''/\epsilon'$ (ratio of loss to storage)

Q: What is μ_r in terms of χ_m ?

$\mu_r = 1 + \chi_m$

Q: Susceptibility of diamagnetic material?

Small and NEGATIVE ($\chi_m < 0$, $\sim 10^{-5}$)

Q: Curie–Weiss law for ferromagnetics?

$\chi_m = C/(T - T_c)$, valid for $T > T_c$

Q: What is coercivity (H_c)?

Reverse H required to bring B back to zero

Q: Example of ferrimagnetic material?

Fe_3O_4 (Magnetite), MnZn ferrite, Ferrites

Q: What is Bloch wall / domain wall?

Boundary between domains; spins gradually rotate across it

Q: What happens to ferromagnet above Curie temp.?

Becomes paramagnetic; domains disappear

§14 Exam Tips & Common Mistakes to Avoid



Confusing ϵ_r and χ_e :

Students write $\epsilon_r = \chi_e$. Wrong! The correct relation is $\epsilon_r = 1 + \chi_e$. For vacuum, $\chi_e = 0$ and $\epsilon_r = 1$. Similarly for magnetics: $\mu_r = 1 + \chi_m$.



Missing vector/tensor signs in $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$:

All three quantities $\mathbf{D}$, $\mathbf{E}$, $\mathbf{P}$ are *vectors*. Never write them as scalars in derivations. The full form is

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$



Forgetting Clausius–Mossotti formula or miswriting it:

Standard form: $(\epsilon_r - 1)/(\epsilon_r + 2) = N\alpha/(3\epsilon_0)$. Never write $(\epsilon_r + 1)/(\epsilon_r - 2)$. The -1 and $+2$ are fixed. At optical frequencies substitute n^2 for ϵ_r .



Errors in polarization type identification:

Order (fastest $\rightarrow$ slowest): *Electronic* $\rightarrow$ *Ionic* $\rightarrow$ *Orientation* $\rightarrow$ *Space charge*. Orientation polarization requires permanent dipoles and is the only one that strongly depends on temperature. Don't say ionic polarization is temperature dependent.



Misunderstanding the hysteresis loop:

Coercivity (H_c) is on the H -axis (where $B=0$). Retentivity (B_r) is on the B -axis (where $H=0$). Common mistake: swapping these. Also: area of loop = energy loss per unit volume per cycle (NOT per second — multiply by frequency for power loss).



Curie vs Curie–Weiss confusion:

Curie law $\chi = C/T$ applies to *paramagnetics*. Curie–Weiss law $\chi = C/(T - T_c)$ applies to *ferromagnetics above* T_c . Below T_c , ferromagnets do NOT follow either law.



Dielectric constant and complex permittivity:

$\epsilon^* = \epsilon' - j\epsilon''$ (note the *minus* sign). ϵ' is the storage (real) part. ϵ'' is the loss (imaginary) part. A high loss tangent $\tan \delta = \epsilon''/\epsilon'$ means the material is lossy (bad for capacitors, good for microwave heating).



Diamagnetic vs Paramagnetic susceptibility signs:

Diamagnetic: χ_m is *negative* (repelled from field). Paramagnetic: χ_m is small *positive* (attracted weakly). Ferromagnetic: χ_m is large *positive* and non-linear. Never assign positive susceptibility to diamagnets.



Ferrimagnetic $\neq$ Antiferromagnetic:

Both have antiparallel dipoles. Key difference: In antiferromagnetics, adjacent antiparallel moments are equal $\rightarrow$ net $M = 0$. In ferrimagnetics, antiparallel moments are unequal $\rightarrow$ net $M \neq 0$. Ferrimagnetics (ferrites) are useful because they have net magnetization.



Soft magnet applications (transformers):

Soft magnets need high permeability (easy magnetization) AND high resistivity (low eddy current loss). Silicon steel (Fe+3-4% Si) increases resistivity without much loss of magnetic properties. Pure Fe has higher permeability but high eddy current loss.



Sure-Shot Exam Formulas (Memorize These!)

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

$$\epsilon_r = 1 + \chi_e$$

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M})$$

$$\mathbf{M} = \chi_m \mathbf{H}$$

$$\mu_r = 1 + \chi_m$$

$$(\epsilon_r - 1)/(\epsilon_r + 2) = Nd/3\epsilon_0$$

$$E_{\text{local}} = E + P/3\epsilon_0$$

$$\chi = C/T \quad (\text{Curie})$$

$$\chi = C/(T - T_c) \quad (\text{Curie-Weiss})$$

§15

Master Formula Quick Reference

No explanations. Pure formulas. Scan and recall.

⚡ ELECTRIC FIELD RELATIONS

$$D = \epsilon_0 E + P$$

$$P = \epsilon_0 \chi_e E$$

$$D = \epsilon_0 \epsilon_r E$$

$$\epsilon_r = 1 + \chi_e$$

$$\epsilon = \epsilon_0 \epsilon_r$$

$$\chi_e = \epsilon_r - 1$$

📡 POLARIZABILITY

$$p = \alpha \epsilon_0 E \quad \text{or} \quad p = \alpha E$$

$$\alpha_e = 4\pi\epsilon_0 R^3$$

$$\alpha_i = e^2 / \omega_0^2 (1/M^+ + 1/M^-)$$

$$\alpha_{or} = p_0^2 / (3k_B T)$$

$$P = N \alpha \epsilon_0 E$$

$$\alpha_{total} = \alpha_e + \alpha_i + \alpha_{or}$$

🌐 LORENTZ FIELD

$$E_{loc} = E + P / (3\epsilon_0)$$

$$E_{loc} = (\epsilon_r + 2) / 3 \cdot E$$

$$E_3 = P / (3\epsilon_0)$$

(Lorentz cavity field)

📊 CLAUSIUS-MOSSOTTI

$$(\epsilon_r - 1) / (\epsilon_r + 2) = N \alpha / (3\epsilon_0)$$

$$(n^2 - 1) / (n^2 + 2) = N \alpha / (3\epsilon_0)$$

[optical, $\epsilon_r = n^2$]

Multiple species:

$$(\epsilon_r - 1) / (\epsilon_r + 2) = \sum N_i \alpha_i / 3\epsilon_0$$

📡 COMPLEX PERMITTIVITY

$$\epsilon^* = \epsilon' - j\epsilon''$$

$$\tan \delta = \epsilon'' / \epsilon'$$

$$P_{loss} = \omega \epsilon_0 \epsilon'' E_0^2 / 2$$

 ϵ' → energy stored ϵ'' → energy lost

🧲 MAGNETIC RELATIONS

$$B = \mu_0 (H + M)$$

$$B = \mu_0 \mu_r H$$

$$M = \chi_m H$$

$$\mu_r = 1 + \chi_m$$

$$\mu = \mu_0 \mu_r$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

⚙️ ATOMIC MAGNETISM

$$\mu_B = e\hbar / (2m_e)$$

$$\mu_B = 9.274 \times 10^{-24} \text{ A} \cdot \text{m}^2$$

$$m = IA \quad (\text{dipole moment})$$

Sources: orbital + spin

$$\text{Spin: } \pm \mu_B \quad (s = \pm \frac{1}{2})$$

🌡️ TEMPERATURE LAWS

Curie (para):

$$\chi = C / T$$

Curie-Weiss (ferro):

$$\chi = C / (T - T_c)$$

Weiss field:

$$H_{mol} = \gamma M$$

📄 HYSTERESIS KEY VALUES

 B_s → Saturation B B_r → Retentivity ($H=0$) H_c → Coercivity ($B=0$)

$$\text{Loss/cycle} = \oint B dH$$

= Area of loop

🌟 SPECIAL VALUES TO MEMORIZE

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$\mu_B = 9.274 \times 10^{-24} \text{ A} \cdot \text{m}^2$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$k_B = 1.38 \times 10^{-23} \text{ J/K}$$

$$\hbar = 1.055 \times 10^{-34} \text{ J} \cdot \text{s}$$

Dielectric Quantity	Formula	Magnetic Analogue	Formula
Electric field E	—	Magnetic field H	—
Displacement D	$\epsilon_0 E + P$	Flux density B	$\mu_0 (H + M)$
Polarization P	$\epsilon_0 \chi_e E$	Magnetization M	$\chi_m H$
Susceptibility χ_e	$\epsilon_r - 1$	Susceptibility χ_m	$\mu_r - 1$
Relative permittivity ϵ_r	$1 + \chi_e$	Relative permeability μ_r	$1 + \chi_m$
Permittivity ϵ	$\epsilon_0 \epsilon_r$	Permeability μ	$\mu_0 \mu_r$

🌀

Final Exam Mantra: $D = \epsilon_0 E + P$ | $\epsilon_r = 1 + \chi_e$ | $B = \mu_0 (H + M)$ | $\mu_r = 1 + \chi_m$ | $(\epsilon_r - 1)/(\epsilon_r + 2) = N\alpha/3\epsilon_0$